Lista 4 - Eletromagnetismo

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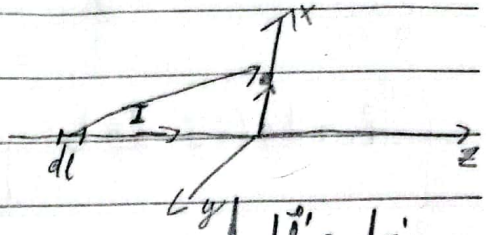
1- $\vec{B} = \oint \frac{\mu_0}{4\pi} \frac{I\, d\vec{l}' \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3}$ → Biot-Savart

$$\vec{B} = \int_{-\infty}^{\infty} \frac{\mu_0}{4\pi} \frac{I\, dz'\hat{a}_z \times (r\hat{a}_r + z'\hat{a}_z)}{|r\hat{a}_r + z\hat{a}_z|^3}$$



$$\vec{B} = \frac{\mu_0 I}{4\pi} \int_{-\infty}^{\infty} \frac{r\, dz'\, \hat{a}_\phi}{(r^2 + z^2)^{3/2}}$$

$d\vec{l}' = dz'$

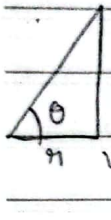
$\vec{x} = r\hat{a}_r$

$\vec{x}' = z\hat{a}_z$

$\hat{a}_z \times (r\hat{a}_r + z'\hat{a}_z)$

$r(\hat{a}_z \times \hat{a}_r) + z(\hat{a}_z \times \hat{a}_z)$

$\boxed{r\, a_\phi}$



$dz' = r\sec^2\theta\, d\theta$

$z' = r\,tg\,\theta$

$tg\,\theta = \frac{z'}{r}$

$$\vec{B} = \frac{\mu_0 I \hat{a}_\phi}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{r^2 \sec^2\theta\, d\theta}{[r^2 + r^2 tg^2\theta]^{3/2}} \Rightarrow \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{r^2 \sec^2\theta\, d\theta}{[r^2(1 + tg^2\theta)]^{3/2}} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cancel{r^2} \sec^2\theta\, d\theta}{\cancel{r^3} \cdot [1 + tg^2\theta]^{3/2}}$$

$$\vec{B} = \frac{\mu_0 I a_\phi}{4\pi r} \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \frac{\sec^2\theta\, d\theta}{(1 + tg^2\theta)^{3/2}}$$

$1 + tg^2\theta = 1 + \frac{sen^2\theta}{cos^2\theta}$

$\frac{cos^2\theta + sen^2\theta}{cos^2\theta} = \frac{1}{cos^2\theta} = \sec^2\theta$

$$\vec{B} = \frac{\mu_0 I a_\phi}{4\pi r} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sec^2\theta\, d\theta}{(\sec^2\theta)^{3/2}} \Rightarrow \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sec^2\theta\, d\theta}{\sec^3}$$

$$\vec{B} = \frac{\mu_0 I a_\phi}{4\pi r} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sec\theta}\, d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\theta\, d\theta = sen\left(\frac{\pi}{2}\right) - sen\left(-\frac{\pi}{2}\right)$$

$1 - (-1) = 2$

$$\vec{B} = \frac{2\mu_0 I \hat{a}_\phi}{4\pi r} = \boxed{\frac{\mu_0 I \hat{a}_\phi}{2\pi r}}$$

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Usando a lei de Ampere temos: $\oint_C \vec{H} d\vec{l} = \int_S \vec{J} \cdot d\vec{S}$

$$\int_0^{2\pi} \vec{H}_\phi \cdot r \, d\phi \, \hat{a}_\phi = \int_0^{2\pi} \int_0^{r} \vec{J} \hat{a}_z \cdot r \, dr \, d\phi \, \hat{a}_z \qquad \vec{J} = \frac{I}{A}$$

$$\int_0^{2\pi} \vec{H} \cdot r \, d\phi (\hat{a}_\phi \cdot \hat{a}_\phi) = \int_0^{2\pi} \int_0^{a} J r \, dr \, d\phi \, (\hat{a}_z \cdot \hat{a}_z)$$

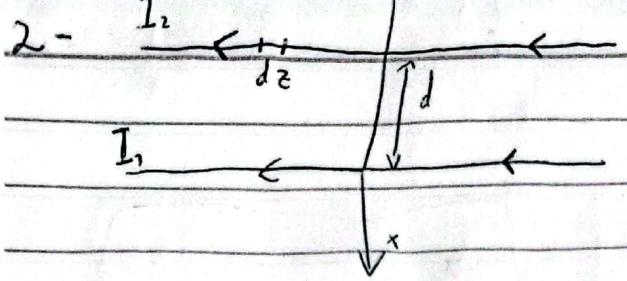
$$H r \int_0^{2\pi} d\phi = \int_0^{2\pi} d\phi \int_0^{a} \frac{I}{\pi a^2} r \, dr$$

$$H r 2\pi = \frac{2\pi I}{\pi a^2} \int_0^{a} r \, dr \Rightarrow \left[\frac{r^2}{2}\right]_0^{a} = \frac{a^2}{2}$$

$$H r 2\pi = \frac{\cancel{2\pi} I \cancel{a^2}}{\cancel{2\pi a^2}} \Rightarrow H = \frac{I}{2\pi r} a_\phi \Rightarrow \vec{B} = \mu_0 \vec{H}$$

$$\boxed{\vec{B} = \frac{\mu_0 I}{2\pi r}}$$

2-



$I_1 = I_2$

$$\vec{F}_{12} = \oint_C I_2\, d\vec{l} \times \vec{B}_1$$

$$\vec{B}_1 = \vec{B}_2 = \frac{\mu_0 I}{2\pi d}\hat{a}_\phi \Rightarrow \vec{F}_{12} = \int_{-\infty}^{\infty} I_2\, dz\, \hat{a}_z \times \frac{\mu_0 I_1 a_\phi}{2\pi r}$$

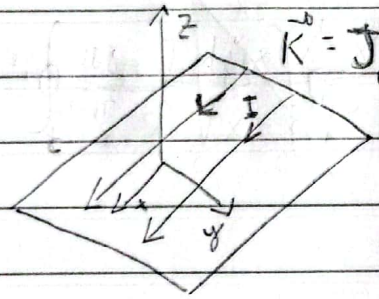
$$\vec{F}_{12} = \int_{-\infty}^{\infty} \frac{I_2\, dz\, I_1 U_0}{2\pi d} \cdot (\hat{a}_z \times a_\phi)$$

$= |\vec{a}_z| \cdot |\vec{a}_\phi| \cdot \operatorname{sen}(\theta)$

$= \hat{a}_r$

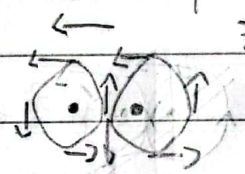
$$\boxed{\vec{F}_{12} = \frac{I_1 I_2 U_0}{2\pi d} \int_{-\infty}^{\infty} dz\, \hat{a}_r}$$

3-

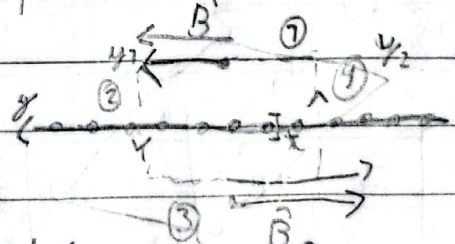


$\vec{K} = J_0 \hat{a}_x$

Vamos utilizar o princípio da superposição com fios infinitos, do ponto de vista $z = 0$, temos o campo se anulando no interior



portanto temos



Usando a Lei de Ampere

$$\oint \vec{H} \cdot d\vec{l} = \int_\Sigma \vec{J} \cdot d\vec{\Sigma}$$

$$\int H\, dy(\hat{a}_y \cdot \hat{a}_y) + \cancel{\int H\, dz(\hat{a}_y \cdot \hat{a}_z)} + \int H\, dy(a_y \cdot a_y) + \cancel{\int H\, dz(\hat{a}_y \cdot \hat{a}_z)}$$

$$\int_{y_1}^{y_2} H(z)\, dy + \int_{y_1}^{y_2} H(z)\, dy = \int_{y_1}^{y_2} \vec{J} \cdot d\vec{\Sigma} \Rightarrow \int_{y_1}^{y_2} \vec{K} \cdot dy\, \cancel{(\hat{a}_x \cdot \hat{a}_x)}$$

$$\vec{J} \cdot d\vec{\Sigma} = \underbrace{\vec{J}_t}_{\vec{K}} \cdot dy$$

$$H(z)(y_2 - y_1) + H(y_1 - y_2) = K(y_2 - y_1)$$

$$\cancel{H(-z)\cdot(y_{z2}-y_{z1})} + \cancel{H(z)(y_{z1}-y_{z2})} = \cancel{K_0(y_{z2}-y_{z1})}$$

$$H(-z) - H(z) = K_0$$

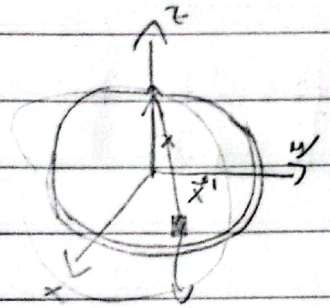
$$\downarrow$$

$$-H(z)$$

$$\Rightarrow -2H(z) = K_0$$

$$H(z)\hat{a}_y = -\frac{K}{2} \Rightarrow \boxed{\vec{H} = -\frac{K}{2}\frac{z}{|z|}\hat{a}_y}$$

4.



$r\,dr\,d\phi$

$$\vec{A} = \int_V \frac{\mu_0}{4\pi}\frac{J(x')\,dV}{|\vec{x}-\vec{x}'|}$$

$$\vec{A} = \frac{\mu_0}{4\pi}\int_0^{2\pi}\int_0^{a}\frac{K_0\hat{a}_x\, r'\,dr'\,d\phi'}{|\vec{z}-\vec{r}'|}$$

$$A = \frac{\mu_0 K_0\hat{a}_x}{4\pi}\int_0^{2\pi}\int_0^{a}\frac{r'\,dr'\,d\phi'}{(z^2-r'^2)^{1/2}} \Rightarrow \cancel{\int_0^{2\pi} d\phi}\int_0^{a}\frac{r'}{(z^2-r'^2)^{1/2}}\,dr'$$

$$2\pi$$

$$\vec{A} = \frac{\mu_0 K_0\hat{a}_x}{2}\int_0^{a}\frac{r'\,dr'}{(z^2-r'^2)^{1/2}}$$

$$u = z^2 - r'^2$$

$$du = 2r'\,dr'$$

$$\frac{du}{2} = r'\,dr'$$

$$\vec{A} = 11\int_{u_1}^{u_2}\frac{du}{2u^{1/2}} \Rightarrow \int_{u_1}^{u_2}\frac{u^{-1/2}}{2}\,du$$

$$\vec{A} = 11\cdot\left[\cancel{2}\frac{u^{1/2}}{\cancel{2}}\right] = u^{1/2}\Big|_{u_1}^{u_2} \Rightarrow \left[z^2-r'^2\right]^{1/2}\Big|_0^a = \sqrt{z^2-a^2} - |z|$$

$$\vec{A} = \frac{\mu_0 K_0\hat{a}_x}{2}\cdot\sqrt{z^2-a^2} - |z|, \text{ para } a \gg z:$$

$$\boxed{\vec{A} = \frac{\mu_0 K_0\hat{a}_x}{2}(|a| - |z|)}$$

$$\vec{B} = \vec{\nabla}\times\vec{A}$$

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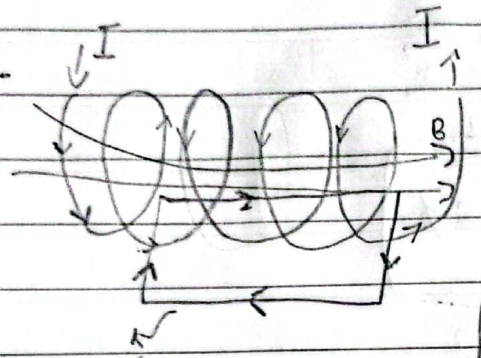
$$\vec{B} = \vec{\nabla} \times \vec{A} = \left(\cancel{\frac{\partial}{\partial x}}^{0} \hat{a}_x + \cancel{\frac{\partial}{\partial y}}^{0} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z \right) \times \frac{(\mu_0 K_0) \cdot (|z| - |z|)}{2} \hat{a}_x$$

$$\vec{B} = \frac{d}{\partial z} \left(\frac{\mu_0 K_0 \cdot [|z| - |z|]}{2} \right) \cdot (\hat{a}_x \times \hat{a}_z) \quad \hat{a}_y$$

$$\boxed{\vec{B} = -\frac{\mu_0 K_0}{2} \hat{a}_y}$$

5-



Vamos usar a Lei de Ampere

$$\oint \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S}$$

$$\int_0^{z_0} H\,dz(\hat{a}_z \cdot \hat{a}_z) + \cancel{\int_0^{H} H\,dr(\hat{a}_r \cdot a_y)}^{0} + \int_{z_0}^{0} H\,dz(\hat{a}_z \cdot \hat{a}_z) + \cancel{\int_{H_0}^{0} H\,dr(-2\pi \cdot \hat{a}_z)}^{0},$$

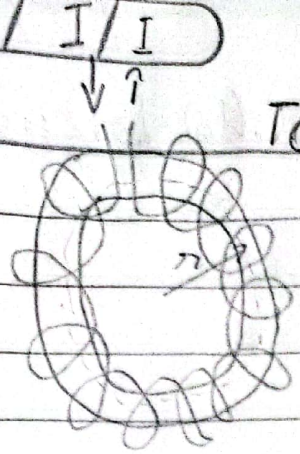
$$\int_0^{z_0} H\,dz - \int_{z_1}^{0} H\,dz = \cancel{\iint_{0\,0}^{z_1\,H_0} \frac{IN}{Lh_0}\,dr\,dz(\hat{a}_\phi \cdot \hat{a}_\phi)}^{77}$$

$$\int_0^{z_1} 2H\,dz = \frac{IN}{L\cancel{h_0}} z_0 \cancel{h_0}$$

$$2H\cancel{z_0} = \frac{IN}{L}\cancel{z_0} \Rightarrow \boxed{\vec{H} = \frac{IN}{2L} \hat{a}_z}$$

$$\boxed{\vec{B} = \vec{H}\mu_0 = \frac{\mu_0 N I}{2L} \hat{a}_z}$$

6- TOROIDE



$$\oint H \cdot dl = \int_S J \cdot dS$$

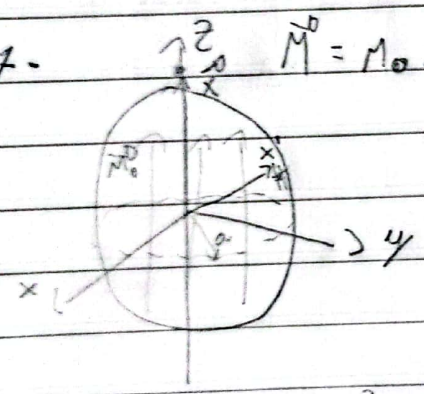
$$\int_0^{2\pi} H r \, d\phi = \int_0^{2\pi}\int_0^{r} \frac{IN}{\pi a^2} r \, dr \, d\phi$$

$$2\pi r H = \frac{2\pi N I}{\pi a^2}\int_0^{r} r \, dr \Rightarrow \frac{r^2}{2}\Big|_0^{a} \quad \frac{a^2}{2}$$

$$\cancel{2\pi} r H = \frac{\cancel{2\pi} N I}{\pi \cancel{a^2}} \cdot \frac{\cancel{a^2}}{2}$$

$$\boxed{\vec{H} = \frac{NI}{2\pi r}\hat{a}_\phi} \qquad \boxed{\vec{B} = \mu_0 \vec{H} = \frac{\mu_0 N I}{2\pi r}\hat{a}_\phi}$$

7-



$\vec{M} = M_0 \hat{a}_z$ $\qquad \vec{B} = \vec{\nabla} \times \vec{A}$

$$\vec{A} = \frac{\mu_0}{4\pi}\int_V \frac{\vec{M}(\vec{x}') \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} dv$$

$\vec{x} = z\hat{a}_z$

$\vec{x}' = R\hat{a}_R$

$\vec{M} = M_0 \hat{a}_z$

$dv = R^2 \sin\theta \, dR \, d\theta \, d\phi$

$$\vec{A} = \frac{\mu_0}{4\pi}\int_0^{a}\int_0^{\pi}\int_0^{2\pi} \frac{M_0 \hat{a}_z \times (z\hat{a}_z - R\hat{a}_R)}{(z^2 + R^2 - 2Rz\cos\theta)^{3/2}} R^2 \sin\theta \, dR \, d\theta \, d\phi$$

$\hat{a}_z \times (z\hat{a}_z - R\hat{a}_R)$

$z(\hat{a}_z \times \hat{a}_z) - R(\hat{a}_R \times \hat{a}_z)$

$0 \qquad \hat{a}_n$

$$\vec{A} = \frac{\mu_0 M_0}{4\pi}\int_0^{a}\int_0^{\pi}\int_0^{2\pi} \frac{R^3 \sin\theta \, dR \, d\theta \, d\phi}{(z^2 + R^2 - 2zR\cos\theta)^{3/2}} \Rightarrow \frac{\cancel{2\pi}\mu_0 M_0}{\cancel{4\pi}}\int_0^{a}\int_0^{\pi} \frac{R^3 \sin\theta \, dR \, d\theta \, \hat{a}_n}{(z^2 + R^2 - 2Rz\cos\theta)^{3/2}}$$

$$\vec{A} = \frac{\mu_0 M_0 \hat{a}_n}{2}\int_0^{a} R^3 dR \int_0^{\pi} \frac{\sin\theta \, d\theta}{(z^2 + R^2 - 2Rz\cos\theta)^{3/2}}$$

$u = z^2 + R^2 - 2zR\cos\theta$

$du = 2zR\sin\theta \, d\theta$

$\sin\theta \, d\theta = \frac{du}{2zR}$

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$$\vec{A} = \frac{\mu_0 M_0 \hat{a}_n}{2} \int_0^a R^3 dR \int_0^{\pi} \frac{\operatorname{sen}\theta \, d\theta}{(z^2+R^2-2zR\cos\theta)^{3/2}}$$

$u = z^2+R^2-2RZ\cos\theta$

$du = 2RZ\operatorname{sen}\theta$

$\operatorname{sen}\theta d\theta = \frac{du}{2RZ}$

$$\vec{A} = \frac{\mu_0 M_0 \hat{a}_n}{2} \cdot \int_0^a R^3 dR \left(\frac{1}{2zR} \int_{u_1}^{u_2} u^{-3/2} du \right)$$

$$\vec{A} = \frac{\mu_0 M_0 \hat{a}_n}{2} \cdot \int_0^a R^3 dR \left(\frac{1}{\cancel{2}zR} \left[\frac{\cancel{2}}{5} u^{-5/2} \right]_{u_1}^{u_2} \right) \Rightarrow \frac{1}{5zR} \cdot \left[\sqrt{(z+R)^2}^{\,5} - \sqrt{(z-R)^2}^{\,5} \right]$$

$$A = \frac{\mu_0 M_0 \hat{a}_n}{2} \int_0^a \frac{R^2 dR}{5zR} \cdot |z+R|^5 - |z-R|^5 \Rightarrow \frac{\mu_0 M_0 \hat{a}_n}{10z} \int_0^a R^2 dR \cdot |z+R|^5 + |z-R|^5$$

Para $z > R = zR^2$

$z < R = 2zR$

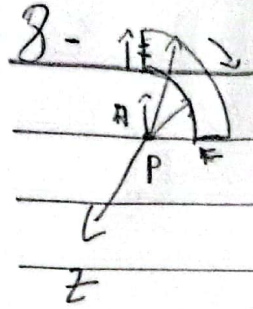
$$\vec{A} = \frac{\mu_0 M_0 \hat{a}_n}{10z} \left(\int_0^z 2R^2 dR + \int_z^a 2Rz dR \right) \Rightarrow \frac{2z^3}{3} + z(a^2 - z^2)$$

$$\vec{A} = \frac{\mu_0 M_0 \hat{a}_n}{10\cancel{z}} \cdot \left(\frac{2z^{\cancel{3}}}{3} + \cancel{z}(a^2 - z^2) \right) = \frac{\mu_0 M_0 \hat{a}_n}{10} \left(\frac{2z^2 + 3a^2 - 3z^2}{3} \right)$$

$$\vec{A} = \frac{\mu_0 M_0 \hat{a}_n}{30} (3R^2 - z^2) \quad \Big\| \quad \vec{B} = \vec{\nabla} \times \vec{A}$$

$$\vec{B} = \left(\frac{\partial}{\partial R} \hat{a}_R + \underbrace{\cancel{\frac{1}{R} \frac{\partial}{\partial \theta}}}_{0} + \underbrace{\cancel{\frac{1}{R\operatorname{sen}\theta} \frac{\partial}{\partial \theta}}}_{0} \hat{a}_\phi \right) \times \frac{\mu_0 M_0 (3R^2 - z^2)}{30} \hat{a}_n$$

$$\vec{B} = \frac{\mu_0 M_0 (3R^2 - z^2)}{30} \frac{\partial}{\partial R} (\hat{a}_R \times \hat{a}_n) = \boxed{\frac{\mu_0 M_0 R \hat{a}_\phi}{10}}$$

8- 

$$\vec{B} = \frac{\mu_0}{4\pi}\oint \frac{I\,d\vec{l}\times(\vec{x}-\vec{x}')}{|\vec{x}-\vec{x}'|^3}$$

$$\vec{B} = \frac{\mu_0}{4\pi}\left(\cancel{\int_a^b \frac{I\,dr\,\hat{a}_r\times(0-r\hat{a}_r)}{|0-r\hat{a}_r|^3}}^{\,0} + \int_{\frac{\pi}{2}}^{0}\frac{I\,b\,d\phi\,\hat{a}_\phi\times(r\hat{a}_r)}{|r\hat{a}_r|^3} + \cancel{\int_b^a \frac{I\,dr\,\hat{a}_r\times(r\hat{a}_r)}{|r\hat{a}_r|^3}}^{\,0} + \int_0^{\frac{\pi}{2}}\frac{I\,a\,d\phi\times(r\hat{a}_r)}{(r\hat{a}_r)^3}\right)$$

$$\vec{B} = \frac{\mu_0 I}{4\pi}\left(\int_{\frac{\pi}{2}}^{0}\frac{-b\,r\,d\phi\,(\cancel{\hat{a}_\phi\times\hat{a}_r})^{-\hat{a}_z}}{b^3} + \int_0^{\frac{\pi}{2}}\frac{a\,r\,d\phi\,(\cancel{\hat{a}_\phi\times\hat{a}_r})^{\hat{a}_z}}{a^3}\right)$$

$$\vec{B} = \frac{\mu_0 I}{4\pi}\left[\frac{1}{b^2}\cdot\frac{r\pi b}{2}\right] + \left[\frac{1}{a^2}\cdot\frac{r\pi a}{2}\right] \Rightarrow \left(\frac{\pi r}{2b} + \frac{\pi r}{2a}\right)\cdot\frac{a}{a}\cdot\frac{b}{b}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi}\cdot\frac{\pi r(a+b)}{2ab}\,\hat{a}_z \Rightarrow \boxed{\frac{\mu I r(a+b)}{8ab}\,\hat{a}_z}$$

9- Lei de Ampere

$$\oint_C \vec{H}\cdot dl = \int_\Sigma \vec{J}\cdot d\vec{\Sigma}$$

Stokes diz

$$\oint_C \vec{H}\,dl = \int_\Sigma \vec{\nabla}\times\vec{H}\,d\Sigma$$

logo temos

$$\frac{d}{d\Sigma}\left(\int_\Sigma \vec{\nabla}\times\vec{H}\,d\Sigma = \int_\Sigma \vec{J}\cdot d\vec{\Sigma}\right)$$

$$\boxed{\vec{\nabla}\times\vec{H} = \vec{J} \rightarrow \text{Equação diferencial}}$$

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10 -

Em um Condutor há um "gap" quase inexistente, com isso, os eletrons na banda de condução podem se mover livremente.

Em um Semicondutor possui um "gap" moderado, sendo assim os eletrons só podem se mover apartir de certas condições como por exemplo o aumento da temperatura ou uma energia

Já o isolante possui uma banda de "gap" significativa, que não permite ao eletrons a possibilidade de se moverem.

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